

SMART TRAFFIC SIGNAL CONTROL USING DEEP Q-NETWORK (DQN)

Full Presenter Script — Word-for-Word Speaking Notes, All 22 Slides

Shaun Joseph Sabu (2362159) · Anantha Narayanan A (2362032) · Delna Liz Denny (2362060) — ~5 minutes 30 seconds total

SECTION 1 · SLIDES 1–3

TEAM

~45 seconds

1 TITLE

SPEAKER 1

Good morning everyone. Today we are presenting our project, Smart Traffic Signal Control Using Deep Q-Network, or DQN.

Our goal is to develop an adaptive traffic signal system using reinforcement learning, instead of relying only on fixed signal timings.

The system focuses on three objectives: emergency vehicle prioritization, fuel and emission reduction with violation monitoring, and traffic congestion reduction.

2 PROBLEM & MOTIVATION

SPEAKER 1

Traffic conditions are constantly changing, but conventional fixed-time signals follow predefined schedules.

This can result in long queues, increased waiting time, unnecessary vehicle stops, fuel consumption and emissions.

Emergency vehicles can also be delayed at red signals, while traffic violations may not be considered by the controller.

Therefore, we need a system that can observe traffic conditions and dynamically adapt the signal.

3 PROPOSED APPROACH & WHY DQN

SPEAKER 1

We model traffic signal control as a Reinforcement Learning problem.

The agent observes the traffic state, selects a signal action, receives a reward, and learns from repeated interactions.

We use DQN because traffic signal actions are discrete, while the traffic state can be complex.

The system can be trained in a simulator such as SUMO.

→ Now, Anantha will explain the first objective — emergency vehicle prioritization.

4 EMERGENCY VEHICLE PRIORITIZATION

The first objective is emergency vehicle prioritization.

Our agent detects approaching emergency vehicles and prioritizes their movement while maintaining efficient traffic flow for other road users.

The important point is that we don't simply prioritize emergency vehicles at any cost. The agent must balance emergency response, normal traffic efficiency and safety.

5 THE PROBLEM

Fixed-time signals may keep an ambulance waiting at a red signal, increasing its response time.

However, immediately stopping other traffic can create large queues elsewhere.

So the challenge is to prioritize the emergency vehicle without unnecessarily disrupting normal traffic.

The signal decision also needs to respect safety constraints such as minimum green time and safe transitions.

6 RL FORMULATION

The agent observes vehicle counts, queue lengths, signal phase and emergency-specific information such as vehicle presence, distance, direction and waiting time.

It can keep the current phase, extend green, switch the phase or activate emergency priority.

The reward reduces emergency waiting time, rewards successful priority and penalizes excessive delay to general traffic and unsafe switching.

7 WORKING SCENARIO

For example, an ambulance approaches from the North while North-South is red and East-West is green.

The system detects the ambulance, identifies its direction, evaluates the possible actions and safely provides priority.

Once the ambulance clears the intersection, the controller returns to adaptive control.

8 OUTCOME

The expected outcome is reduced emergency waiting time, faster emergency movement and improved response efficiency, while minimizing disruption to normal traffic.

These results still need to be experimentally evaluated against a fixed-time baseline.

→ Now I'll hand over to Shaun.

9 FUEL, EMISSIONS & VIOLATIONS

The second objective focuses on reducing unnecessary stops, fuel consumption and vehicular emissions while monitoring traffic violations.

10 THE PROBLEM

Repeated stopping, idling and acceleration create inefficient traffic movement and can increase fuel consumption and emissions.

Our goal is therefore to achieve smoother traffic progression with fewer unnecessary stops.

11 RL FORMULATION

The agent considers vehicle count, queue length, waiting time, speed, number of stops, estimated emissions and violation indicators.

It can keep, extend, switch or reduce the green phase.

The reward encourages smooth movement and fewer stops, while penalizing increasing queues, emissions and unsafe behaviour.

12 VIOLATION DETECTION

The system also monitors red-light violations, illegal crossings, lane violations and unsafe signal-related behaviour.

These events are represented as violation indicators and can contribute to the reward penalty and safety metrics.

The case study does not specify a particular computer-vision model for this component.

13 OUTCOME

The expected result is fewer unnecessary stops, lower idle time, lower estimated fuel consumption and emissions, smoother traffic and violation monitoring.

→ Now Delna will explain our third objective — congestion reduction.

14 CONGESTION REDUCTION

The third objective is to reduce traffic congestion by minimizing queue lengths and waiting times through adaptive signal control.

The key idea is that green time should respond to actual traffic demand.

15 THE PROBLEM

For example, if Road A has 30 vehicles while Road B has only 5, a fixed-time signal may still allocate predetermined green times.

This can create unnecessary queues and poor green-time utilization.

An adaptive agent can instead respond to where the traffic demand actually exists.

16 RL FORMULATION

The state includes queue lengths, vehicle count, waiting time, arrival rate, signal phase, phase duration and average speed.

The agent can keep, extend, switch or reduce the green phase.

The reward is: negative queue length, negative waiting time, plus throughput.

So lower queues and waiting, together with higher throughput, produce better rewards.

17 WORKING SCENARIO

If North-South has a large queue and East-West has few vehicles, the DQN can detect this imbalance and extend North-South green.

The queue decreases, producing a better reward, and the experience contributes to learning.

18 OUTCOME

The expected outcomes are lower queue lengths, lower waiting time, higher throughput and better green-time utilization.

In simple terms: Green time follows demand, not the clock.

→ Now we'll bring all three objectives together.

19 THREE OBJECTIVES, ONE SYSTEM**SPEAKER 1**

Now we combine the three objectives into one framework.

Anantha's component focuses on emergency priority, Shaun's on fuel, emissions and violations, and Delna's on congestion reduction.

These are not three completely independent systems.

They contribute to one unified DQN traffic signal controller, with a common state representation, action space, multi-objective reward and learned policy.

20 MULTI-OBJECTIVE REWARD**SPEAKER 1**

The overall reward combines queue length, waiting time, unnecessary stops, emissions, emergency priority and violation penalties.

This allows the agent to balance competing objectives instead of optimizing only one aspect of traffic. For example, emergency priority may temporarily increase normal traffic delay, so the reward helps the agent learn an appropriate trade-off.

21 DQN WORKFLOW & EVALUATION**SPEAKER 2**

The complete learning cycle is: State → DQN → Action → Traffic Environment → Reward → Next State → Learning.

During training, experiences are stored in replay memory and used to update the DQN.

We evaluate the system using traffic metrics such as waiting time, queue length and throughput, environmental metrics such as fuel and emissions, emergency-response metrics, and safety metrics such as violations.

These will be compared against a conventional fixed-time or rule-based baseline.

SPEAKER 3

Our key direction is an explainable, multi-objective DQN that can make decisions while providing understandable reasons for them.

Future extensions can include Multi-Agent Reinforcement Learning for coordinated intersections.

The major challenges include reward design, training time, exploration, generalization, safety and the simulation-to-real-world gap.

To conclude, our three objectives target emergency response, environmental efficiency and congestion reduction.

Together, they form:

“Three Objectives. One Intelligent DQN-Based Traffic Control Framework.”

Thank you.